

OBONE ESSONO Hegirziana Heva

Seeking a 4 to 6-month Internship – Starting February 2027

Master's Degree in Medical Radiation Physics

📍 Toulouse, France

☎ +33 6 68 82 65 82

✉ essono.heva@gmail.com

🌐 [essonoheva-git](https://github.com/essonoheva)



Education & Qualifications

- **M.Sc. in Health Engineering - Medical Radiation Physics** Toulouse, France
Faculty of Science and Engineering 2026 - Present
 - Dosimetry, Medical Imaging Techniques (**MRI, PET, SPECT, Ultrasound**), Radiotherapy, Radiobiology, Radiation-Matter Interactions, Computer Science Techniques.
- **Bachelor's Degree in Science, Technology, and Health** Toulouse, France
Faculty of Science and Engineering 2025

Professional Experience

- **Medical Physics Intern** Libreville, Gabon
Akanda Cancer Institute (ICA) 22/06/2026 - 31/07/2026
 - **Topic:** Immersion within the **Radiotherapy** and **Nuclear Medicine** departments.
 - Performed daily, weekly, and monthly quality assurance (QA) tests on radiotherapy equipment (**LINAC, CT Simulator, ISOMED dose calibrator**).
 - Introduction to the **Treatment Planning System (TPS)** (delineation of organs at risk, contouring of treatment volumes (CTV, PTV), field creation, selection of particle types and energies).
 - Authored a comprehensive internship report summarizing clinical procedures and best practices.
- **Research Intern** Toulouse, France
LAPLACE Laboratory — PRHE Group 23/06/2025 - 18/07/2025
 - **Topic:** Experimental study of cold plasma / microorganism interactions.
 - Participated in **plasma physics** research investigating the efficacy of **reactive species (ROS/RNS)** generated by plasma discharge for biological decontamination (sterilization).

Academic Projects

- **Master 1 Project - Brain Metabolic Quantification** Toulouse, France
Toulouse NeuroImaging Center (ToNIC) Mar 2026 - May 2026
 - **Topic:** **PET Quantification** of brain biomarkers and analysis of regional metabolic deficits in Alzheimer's disease.
 - Preprocessing of 3D brain volumes using SPM (spatial coregistration, normalization, and resampling via trilinear interpolation), **SUV and SUVR calculation** using the cerebellum as the reference region.
 - Quantitative metabolic analysis in key disease regions (frontal cortex, precuneus, posterior cingulate) comparing profiles (healthy subjects vs. cognitive impairment / Alzheimer's disease).
- **Master 1 Project - Medical Image Viewer Interface in C++** Toulouse, France
Faculty of Science and Engineering Feb 2026 - May 2026
 - **Topic:** Design of an interactive 3D visualization and processing interface for medical images (**DICOM CT data**).
 - Implementation of a 2D Gaussian filter via spatial convolution and Look-Up Tables (LUT) for contrast adjustment and color mapping (Jet Palette).
 - Simulation of a **radiotherapy beam** (dose penetration management) and reconstruction of **Digitally Reconstructed Radiographs (DRR)**.

- Segmentation of anatomical structures (bone/tissue) and fluid real-time rendering integrated into the Graphical User Interface (GUI).

- **Bachelor's Project - Vectorization & Huntington's Disease**
SOFTMAT Laboratory

Toulouse, France
Sep 2024 - Jan 2025

- **Topic:** Study of **vectorization techniques** for targeted drug delivery in the treatment of neurodegenerative diseases.

Neuroinformatics & Medical Imaging

- **Software:** SPM12 (Statistical Parametric Mapping), MRICron/MRicroGL
- **Analysis:** PET/MRI image processing, SUVR calculation, spatial normalization (MNI space)

Programming Skills

- **Languages:** C, C++, MATLAB
- **Development Platforms:** Visual Studio Code, PyCharm, Qt Creator, Ubuntu
- **Operating Systems:** Linux, Windows, Android

Core Competencies & Soft Skills

- Scientific rigor and strict adherence to protocols
- Analytical skills and synthesis capability
- Teamwork and interdisciplinary communication
- Autonomy and adaptability

Languages

- **French:** Native
- **English:** C1 (Fluent)
- **Spanish:** B1 (Intermediate)
- **Mandarin:** A2 (Elementary)

Honors & Awards

- **Merit-based Scholarship:** Awarded for High Honors on the High School Diploma (Baccalauréat - Mention Très Bien)